# UTHEALTH HOUSTON AND MEMORIAL HERMANN HIGH RISK LIVER TRANSPLANT PROTOCOL FOR NEUROENDOCRINE LIVER METASTASES V7

Determine whether LT offers meaningful additional benefit over contemporary NET treatment, then distinguish national exception guidance from local prognostic selection and true experimental expansion. Indolent disease with excellent expected non-LT survival must not be transplanted merely to produce favorable recipient survival.

Version 7 | Revised 2026-10-24 06:55 CDT | Institutional approval pending

## POLICY AND ALLOCATION STATUS

Analyst A structured critical review through 24 October 2026. All 27 B decisions remain OPEN. Primary evidence, selected-recipient outcomes, pathway-level outcomes, guideline wording and local safeguards are distinguished. This is not a PRISMA-complete systematic review, meta-analysis, fresh external-analyst consensus or institutional approval.

Policy posting checked 24 October 2026: OPTN effective 1 October 2026; oncology guidance December 2025. Clinical suitability, allocation access, donor protection and research authorization remain separate. Recheck before any application. [1] [2]

Use the adjacent study tables before interpreting a proposed cutoff. Abstract-only extraction and missing full PDFs are labeled; missing outcome estimates are not inferred from graphs or from one minus disease-free survival. Linked full text is in this document's own PDF depot when available.

## 1 PURPOSE AND EVIDENCE BASIS

The Milan-NET comparison included 42 recipients and 46 nontransplanted patients, with substantial age and treatment differences. Five-/ten-year OS was 97.2%/88.8% versus 50.9%/22.4%. It was nonrandomized and predates many current treatment strategies; it does not determine benefit versus contemporary PRRT, targeted therapy or liver-directed care. [3] [4]

The Spanish historical series included 91 recipients, 41 transplanted before 2007. Table 5 reported five-year OS 71.0% within Milan versus 58.7% outside Milan, with an OS difference; the nonsignificant DFS comparison did not show equivalence. There were 52 recurrences, nine 90-day deaths and nine reported retransplant events: two primary nonfunction, six hepatic-artery thrombosis and one late rejection. These are historical recipient data, not modern center-specific risks or a non-LT comparison. [5]

## NET-B1 — WHICH PATIENTS GAIN ENOUGH OVER MODERN NON-LT TREATMENT?

### QUESTION AND COMPARISON

Among technically eligible liver-dominant GEP-NET patients, who is likely to gain enough from LT compared with feasible contemporary non-LT care, rather than simply survive well after LT?

## EVIDENCE BEFORE CHOOSING A RULE

| Study • population • source | Results • interpretation boundary |
|---|---|
| **Mazzaferro 2016 · Milan-NET** [3]<br>Single-centre comparative Milan-NET study, 88 eligible patients: 42 LT and 46 non-LT, 1995–2010 era, nonrandomized.<br>Primary source \| Full-text PDF<br>Locator: Am J Transplant 2016;16:2892–2902; full PDF Methods statistical time origin/left truncation, Tables 1–2 eligibility/characteristics, survival and restricted-mean analyses.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Median follow-up 122 months. Five-/ten-year OS 97.2%/88.8% LT versus 50.9%/22.4% non-LT. Time origin was primary-tumour resection; authors used left-truncated Cox models to account for delayed LT entry. Propensity-adjusted restricted mean survival gain was 6.82 months at five years and 38.43 months at ten years. These are cohort estimates, not individual calculators.<br>**Do not infer:** Major selection/age/treatment-era differences, small numbers and residual confounding remain despite statistical handling of delayed treatment. Only 4/46 non-LT patients received PRRT, so this is not a modern sequential-care comparator. Patients overlap subsequent Milan series and some multicentre LT studies; do not sum them as independent evidence. |
| **Eshmuminov 2023 · LT versus resection** [6]<br>International retrospective comparison from 15 centres, 1988–2021: 455 patients (225 LT, 230 liver resection); median follow-up 97 months. Propensity matching produced 96 patients per group.<br>Primary source \| Full-text PDF<br>Locator: Ann Surg 2023 DOI10.1097/SLA.0000000000005663; PDF main Results printed p.e1067/Table 4 for matched outcomes; earlier Tables 1–3 for grade and mortality; abstract on first page differs.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Use main Results p.e1067/Table 4: matched five-year OS 75% LT versus 68.3% LR, HR 0.56 (95% CI 0.35–0.90); five-year PFS 64.2% versus 14.2%, HR 0.25 (0.16–0.39), measured from operation. Ninety-day deaths 13/225 LT (5.8%) versus 3/230 LR (1.3%). LT G1 versus G2 five-year OS 92.9% versus 70%; adjusted G2 HR 2.52 (1.15–5.52). Outside-Milan matched OS HR 0.64 (0.31–1.33), not statistically conclusive.<br>**Do not infer:** Material abstract/body discrepancy: abstract reports matched five-year OS 73% versus 52.8%, whereas main Results/Table 4 report 75% versus 68.3%. The latter is used with explicit discrepancy flag; clarification would be desirable. Matching does not remove unmeasured confounding, missing pathology, era or selection bias. The nonsignificant outside-Milan estimate does not prove no benefit, and ten G3 cases do not validate G3 LT. Overlaps centre cohorts including Milan. |

| Study • population • source | Results • interpretation boundary |
| --- | --- |
| **Maspero 2022 · Milan long-term** [7]<br>Retrospective Milan single-centre series, 1984–2019: 149 liver operations (53 LT, 96 LR), with 104 Milan-eligible cases analysed (48 LT, 56 LR).<br>Primary source \| Full-text PDF<br>Locator: Am J Transplant 2022 DOI10.1111/ajt.17156; complete PDF pp.2–4 methods/time origin/cohort; later survival figures, recurrence table and Discussion.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Time zero liver operation; median follow-up 158 months LT and 126 LR. Five-year OS 95.5% versus 90%; ten-year OS 93% versus 75%. Ten-year DFS 52% versus 18%; recurrence 25/48 versus 42/56 and median recurrence time 78 versus 24 months. LT recurrence was more often multisite; LR recurrence predominantly liver. Nine LT deaths included three cancer and six non-cancer deaths, including new malignancy, sepsis/lymphoproliferative and cardiovascular causes.<br>**Do not infer:** Not randomized or anatomically exchangeable; prolonged selected follow-up and differing age can favor LT. Strong overlap with 2016 Milan-NET makes it long-term/alternative-comparator information, not independent confirmation. Late recurrence and immunosuppression-related competing causes matter even with excellent survival. |
| **Kjaer 2023 · eligible non-LT panNET** [8]<br>Single-centre retrospective non-LT panNET cohort, 2000–2021: 519 screened, 41 G1/G2 metastatic patients included; 16 met at least one historical LT framework (Milan 11, ENETS 8, UNOS 13, overlapping).<br>Primary source \| Full-text PDF<br>Locator: World J Surg 2023 DOI10.1007/s00 268-022-06 736-1; complete PDF pp.2–5 methods, time origin, eligibility modifications, treatments and survival tables.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Time zero was date LT criteria fulfilled (for remaining patients, initial institutional visit after primary resection with liver metastases). Mean follow-up 5.4 years. Overall 41-patient median OS 9.3 years; five-year OS 64.7% (95% CI 48.2–81.2). Milan subset five-year OS 64.9% (32.2–97.6), ENETS 85.7% (59.8–100), historical UNOS 55.4% (26.0–84.8).<br>**Do not infer:** Small retrospective selected subgroups with very wide intervals; no internal LT comparator and treatment-era limitations. It challenges the assumption that eligible stage IV NET has uniformly poor non-LT survival; it cannot prove that non-LT is superior to LT. Avoid comparing its eligibility-based survival directly with post-LT curves as if randomized. |

| Study • population • source | Results • interpretation boundary |
| --- | --- |
| **Chan 2025 · CABINET** [9]<br>CABINET double-blind phase III randomized trial, 298 previously treated progressive well-differentiated NET patients: independent extrapancreatic cohort n=203 and pancreatic cohort n=95, 2:1 cabozantinib versus placebo.<br>Primary source \| Full-text PDF<br>Locator: N Engl J Med 2025;392:653–665, online 2024; local complete PDF Methods, primary Results, response and safety tables.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Time zero randomization. Median PFS 8.4 versus 3.9 months in extrapancreatic NET (HR 0.38, 95% CI 0.25–0.59), 13.8 versus 4.4 months in panNET (HR 0.23, 0.12–0.42). Objective response 5% and 19% in respective cabozantinib cohorts versus 0% placebo. Grade ≥3 adverse events approximately 62–65% versus 23–27%. The primary report does not establish an LT-comparable OS benefit.<br>**Do not infer:** High-quality evidence for this systemic-treatment setting, indirect for transplant benefit. A treatment's availability does not mean every candidate must receive it before LT, and its toxicities/tempo must enter individualized comparison. PFS is not OS and placebo control does not represent an entire modern lifetime sequence. |
| **Caplin 2014 · CLARINET** [10]<br>CLARINET randomized double-blind trial, 204 nonfunctioning advanced enteropancreatic NET patients (101 lanreotide, 103 placebo).<br>Primary source \| Full-text PDF<br>Locator: N Engl J Med 2014;371:224–233; local complete PDF eligibility, Results and adverse-event table.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** From randomization, median PFS not reached with lanreotide versus 18 months placebo; HR 0.47 (95% CI 0.30–0.73). Twenty-four-month PFS 65.1% versus 33.0%. No significant OS or QoL difference in the primary report. Diarrhea 26% versus 9%.<br>**Do not infer:** Shows that stability and substantial survival without LT are plausible; does not estimate the added benefit of transplant or support treating all stable low-burden NET. G2 up to 20% and G3 are not represented by this <10% trial threshold. |

| Study • population • source | Results • interpretation boundary |
|---|---|
| **Strosberg 2021 · NETTER-1 final OS** [11]<br>NETTER-1 phase III randomized trial final OS follow-up: 360 assessed, 231 randomized (117 Lu-177-Dotatate, 114 high-dose octreotide), 223 treated in safety population.<br>Primary source \| Full-text PDF<br>Locator: Lancet Oncol 2021;22:1752–1763; local corrected PDF pp.5–6 participant flow/crossover/OS; safety Results and Discussion.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Time zero randomization; median follow-up about 76 months. Median OS 48.0 versus 36.3 months, HR 0.84 (95% CI 0.60–1.17), P=0.30; primary final OS endpoint not met. Forty-one of 114 controls (36%) crossed to PRRT. Two of 111 Lu-treated safety patients developed myelodysplastic syndrome, one fatal. Previously established PFS benefit and long-term tolerability remain relevant despite nonsignificant OS.<br>**Do not infer:** Crossover, additional therapies, attrition and distinct population prevent using median OS as a transplant counterfactual. Do not equate nonsignificant OS with no clinical activity, or assume PRRT has no long-term marrow risk. |
| **ENETS 2026** [4]<br>ENETS 2026 position statement explicitly addressing controversies; expert, hypothesis-generating framework rather than original comparative outcome study.<br>Primary source \| Full-text PDF<br>Locator: J Neuroendocrinol 2026;38:e70 250; complete position-statement PDF burden, benefit, age, stability and extrahepatic discussion.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** No original outcome denominator. Proposed liver-burden bands around 25–70%, age <60 as relative and six-month stability are pragmatic concepts. Illustrative non-LT five-year survival <30% versus post-LT ≥70% are not a calibrated individual benefit model.<br>**Do not infer:** Do not turn an expert future-oriented proposal into validated numerical gates. Suggested controllable extrahepatic disease, including bone, conflicts with current NLRB solid-organ-metastasis exclusion. It identifies research needs rather than approving an expansion. |

## SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

Selected NET recipients can have excellent long-term survival, and comparative cohorts suggest potential benefit, especially over a decade. However, good post-LT survival is not synonymous with large incremental benefit: non-LT eligible patients can also live many years. The best Milan comparison adjusted delayed LT entry but remains confounded and predates most modern sequencing. The multicentre main-body matched OS difference is smaller than its abstract suggests. A credible benefit assessment must include both long-term hepatic threat and competing treatment/transplant harms. [3] [6] [7] [8] [9] [10] [11] [4]

- Kjaer's eligibility-based non-LT cohort had substantial long-term survival; CLARINET supports prolonged control for indolent Ki-67 <10% disease. CABINET and PRRT offer additional disease-control options, but none is a randomized comparator to LT. Conversely, requiring every systemic option to fail may forfeit the transplant window and is not tested. Maspero documents late multisite recurrence and non-cancer deaths despite excellent LT survival.

**Certainty / applicability:** Low certainty for a causal LT benefit magnitude and any individual benefit cutoff; moderate certainty that NET biology and modern alternatives vary greatly. High-quality systemic-trial efficacy is indirect for transplant selection. No validated calculator supports the ENETS illustrative 30%/70% survival targets.

## WHERE EACH PROPOSED THRESHOLD COMES FROM

- No randomized study validates a minimum survival-gain percentage, two-line failure count, or ENETS <30% versus ≥70% prediction. Six-month observation screens biology but does not prove added benefit. Any local two-reviewer requirement and reassessment cadence are governance, not biological cutoffs.

## SOURCE WORDING — READ IN CONTEXT

| Reference and exact excerpt | What it supports / does not establish |
|---|---|
| **ENETS 2026** [4]<br>"expert-driven and hypothesis-generating framework (not outcome-validated)"<br>2026 full-text PDF p3, transplant-benefit discussion<br>Source \| Full-text PDF | Illustrative survival figures are discussion aids, not calibrated individual eligibility thresholds. |

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

## OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

**Proposed stance:** Require a demonstrable benefit trigger; excellent predicted post-LT survival alone is not enough. [4] [3] [6] [7] [8] [9] [10] [11]

- Proposed gate: favorable-biology unresectable liver disease with controlled primary, plus at least one trigger—refractory functioning syndrome, clinically meaningful liver-related functional compromise, or a prospectively documented credible threat to durable hepatic control despite reasonable alternatives.
  - Do not require established irreversible failure before assessing benefit.
- NET oncology and an independent transplant/HPB reviewer must agree that LT offers meaningful additional survival or symptom/function benefit over the patient's accessible options, with B2 biological stability satisfied.
- Asymptomatic stable low-burden disease with preserved hepatic function and reasonable durable options defaults to continued non-LT care/observation.
  - Evaluate at least one appropriate treatment strategy; do not require two failures or exhaustion of every therapy.
  - Do not use ENETS illustrative <30%/≥70% figures as a validated calculator.

**Why retain or change it:** Retain OPEN an explicit incremental-benefit review by transplant and NET specialists independent of the donor decision. Record current and realistically available sequences (SSA, PRRT if appropriate, targeted/systemic agents, liver-directed therapy/resection), prior response/toxicity, symptom burden, functional liver reserve and credible future threat to hepatic control. A compelling case may be refractory hormonal symptoms, threatened hepatic function, or a carefully documented likely loss of durable liver control despite reasonable alternatives—not necessarily established irreversible failure. Do not require two failed lines or exhaustion of all drugs. Stable, low-burden disease with durable tolerable control ordinarily remains in non-LT management. Decision remains blank.

### INFORMATION REQUIRED BEFORE THE DECISION

Functional syndrome and treatment response; symptoms/liver reserve; growth trajectory; SSTR status and treatment alternatives; patient goals; independent benefit review.

**Clinical versus allocation:** Favorable clinical prognosis does not establish urgency, exception entitlement or a reason to expose a healthy donor.

**Roadmap / protocol reconciliation:** Compare with the NET roadmap's SSA, PRRT, liver-directed, targeted and chemotherapy options as applicable, without forcing inappropriate therapies before LT review.

Meeting record: NET-B1 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

## 2 PATIENT SELECTION AND CLASSIFICATION

### 2.1 PATHOLOGY AND PRIMARY SITE

- Obtain expert review of the primary and metastatic pathology when available: differentiation, grade, Ki-67 hot-spot index, mitotic count, sampling date and heterogeneity. Resolve discordant primary/liver grades or a rapidly changing phenotype with the NET MDT.
- Current NLRB guidance specifies portal-drained gastroenteropancreatic origin, G1/G2 histology, mitotic rate below 20 per 10 HPF and proliferation index below 20%. Do not change the exact index boundary to less-than-or-equal-to 20. Record unknown or nonportal primary sites separately. [2]
- Well-differentiated G3 NET is biologically distinct from poorly differentiated NEC. Neither is activated as a transplant cohort here. NETTER-2 demonstrates non-LT PRRT activity in a selected G2/G3 population, not transplant benefit or a validated Ki-67 below 55 LT gate. [12]

### 2.2 STAGE NODES AND EXTRAHEPATIC DISEASE ACCURATELY

- Obtain multiphasic liver CT/MRI, chest/abdominal/pelvic staging and appropriate somatostatin-receptor functional imaging. Add FDG imaging when grade, rapid tempo or receptor/anatomic discordance makes it clinically useful. Resolve suspicious sites with directed imaging or biopsy where management depends on the result.
- Document complete primary resection and any prior extrahepatic nodal resection, exact stations, pathology and dates. Guidance requires at least six months without recurrence after primary/extrahepatic disease clearance; current extrahepatic progression, including positive nodes, prevents advancement. Previously cleared nodes are not equivalent to active nodes. [2]
- Prior or current extrahepatic solid-organ metastases, such as lung or bone metastases, are an exclusion in the current NET exception guidance. Do not apply the cleared-node reconsideration language to those sites. Record the required negative metastatic-workup dates and obtain current staging before activation rather than relying on an old negative scan. [2]

### 2.3 BURDEN AGE AND CLINICAL BENEFIT

- Measure percentage liver replacement with a specified method, functional reserve, symptoms/hormonal syndrome and treatment trajectory. Current NLRB guidance contains no explicit age or percentage liver-volume cap. Original Milan age/burden selection factors must not be labeled national policy. [2] [3]

- The former 50–75% burden band with mandatory downstaging to 50%, and the age 60–70 research band, are not adopted as validated limits. Assess physiologic fitness and expected incremental benefit; any local restriction or expanded stratum needs an explicit rationale. [13] [4]

## NET-B2 — WHAT IS GUIDELINE-ALIGNED VERSUS AN EXPANDED-RISK PROPOSAL?

### QUESTION AND COMPARISON

What pathology, primary-site, nodal/extrahepatic and stability boundaries should govern NET LT; can G3 or PRRT-responsive disease be promoted into the pathway?

### EVIDENCE BEFORE CHOOSING A RULE

| Study • population • source | Results • interpretation boundary |
|---|---|
| **Mazzaferro 2016 · Milan-NET** [3]<br>Single-centre comparative Milan-NET study, 88 eligible patients: 42 LT and 46 non-LT, 1995–2010 era, nonrandomized.<br>Primary source \| Full-text PDF<br>Locator: Am J Transplant 2016;16:2892–2902; full PDF Methods statistical time origin/left truncation, Tables 1–2 eligibility/characteristics, survival and restricted-mean analyses.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Median follow-up 122 months. Five-/ten-year OS 97.2%/88.8% LT versus 50.9%/22.4% non-LT. Time origin was primary-tumour resection; authors used left-truncated Cox models to account for delayed LT entry. Propensity-adjusted restricted mean survival gain was 6.82 months at five years and 38.43 months at ten years. These are cohort estimates, not individual calculators.<br>**Do not infer:** Major selection/age/treatment-era differences, small numbers and residual confounding remain despite statistical handling of delayed treatment. Only 4/46 non-LT patients received PRRT, so this is not a modern sequential-care comparator. Patients overlap subsequent Milan series and some multicentre LT studies; do not sum them as independent evidence. |

| Study • population • source | Results • interpretation boundary |
| --- | --- |
| **Eshmuminov 2023 · LT versus resection** [6]<br>International retrospective comparison from 15 centres, 1988–2021: 455 patients (225 LT, 230 liver resection); median follow-up 97 months. Propensity matching produced 96 patients per group.<br>Primary source \| Full-text PDF<br>Locator: Ann Surg 2023 DOI10.1097/SLA.0000 000 000 005 663; PDF main Results printed p.e1067/Table 4 for matched outcomes; earlier Tables 1–3 for grade and mortality; abstract on first page differs.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Use main Results p.e1067/Table 4: matched five-year OS 75% LT versus 68.3% LR, HR 0.56 (95% CI 0.35–0.90); five-year PFS 64.2% versus 14.2%, HR 0.25 (0.16–0.39), measured from operation. Ninety-day deaths 13/225 LT (5.8%) versus 3/230 LR (1.3%). LT G1 versus G2 five-year OS 92.9% versus 70%; adjusted G2 HR 2.52 (1.15–5.52). Outside-Milan matched OS HR 0.64 (0.31–1.33), not statistically conclusive.<br>**Do not infer:** Material abstract/body discrepancy: abstract reports matched five-year OS 73% versus 52.8%, whereas main Results/Table 4 report 75% versus 68.3%. The latter is used with explicit discrepancy flag; clarification would be desirable. Matching does not remove unmeasured confounding, missing pathology, era or selection bias. The nonsignificant outside-Milan estimate does not prove no benefit, and ten G3 cases do not validate G3 LT. Overlaps centre cohorts including Milan. |
| **Boscà 2026 · Spain** [5]<br>Spanish 13-centre retrospective series of 91 NET LT recipients, 1995–2024; 41 transplanted before 2007; 71 Milan-IN and 20 Milan-OUT.<br>Primary source \| Full-text PDF<br>Locator: J Clin Med 2026;15:938; complete PDF Results, Tables 2–5, postoperative mortality/retransplantation and limitations.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Time zero LT; median follow-up 6.5 years. Table 5: five-year OS 71.0% IN versus 58.7% OUT, ten-year 55.5% versus 32.0%, fifteen-year 52.2% versus 16.0% (OS comparison P=0.03). Five-year DFS 58.8% versus 36.3%, not statistically significant. Recurrence 52/91; ninety-day mortality 9/91 (9.9%). Nine retransplants: two primary nonfunction, six hepatic artery thrombosis, one rejection.<br>**Do not infer:** Era, missing pathology, small OUT group and composite deviations preclude validating 50–70% burden, age ≥60, G3 or extrahepatic expansion individually. Nonsignificant DFS is not equivalence. The abstract rounds OUT five-year OS differently; use Table 5 value with locator. |

| Study • population • source | Results • interpretation boundary |
|---|---|
| **Le Treut 2013 · ELTR** [14]<br>European Liver Transplant Registry retrospective multicentre study, 213 recipients at 35 centres, 1982–2009; 107 before 2000 and 106 thereafter. Published primary abstract reviewed.<br>Primary source \| TMC: retrieve PDF<br>Locator: Ann Surg 2013 DOI10.1097/SLA.0b013e31 828ee17c; PMID23 532 105 primary abstract; complete PDF not located.<br>Review depth: Published primary abstract reviewed; complete PDF not available in searched depots/Downloads | **Reported result:** Three-month mortality 10%. Five-year OS 52% measured from LT, versus 73% measured from liver-metastasis diagnosis; five-year DFS 30%. Post-2000 five-year OS 59%. Poor differentiation, hepatomegaly and concurrent major resection were adverse; age >45 was adverse in a later-era analysis.<br>**Do not infer:** The striking time-origin difference illustrates why survival curves cannot be compared casually. Registry era, incomplete grading and nonrandom selection limit modern applicability. Age 45 association does not establish a biological exclusion threshold. Cohort overlap with later multicentre reports is likely. |
| **NETTER-2** [12]<br>NETTER-2 open-label phase III randomized trial: 261 screened, 226 randomized, 151 Lu-177-Dotatate plus octreotide and 75 high-dose octreotide.<br>Primary source \| Full-text PDF<br>Locator: Lancet 2024;403:2807–2817; complete PDF eligibility, efficacy, safety and Discussion.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Time zero randomization. Median PFS 22.8 versus 8.5 months, HR 0.276 (95% CI 0.182–0.418); objective response 43% versus 9.3%. OS was not mature. Safety populations 147 and 73: any grade ≥3 events 35% versus 27%, related grade ≥3 events 16% versus 4%; one myelodysplastic syndrome case and three grade-3 nephrotoxicity events in Lu-treated patients; no drug-related deaths reported.<br>**Do not infer:** Compelling systemic disease-control evidence for selected well-differentiated G2/G3, not evidence that responding G3 becomes suitable for LT or that marrow/renal injury is irrelevant to future transplantation. No NEC efficacy extrapolation. Full article reviewed now, upgrading prior abstract-only status. |
| **Chen 2026 · PRRT after LT** [15]<br>Retrospective seven-recipient case series of PRRT for recurrent NET after LT; abstract-only verification.<br>Primary source \| TMC: retrieve PDF<br>Locator: Transplant Proc 2026;58:526–531; DOI10.1016/j.transproceed.2026.02.019, PMID41 760 506 abstract.<br>Review depth: Published primary abstract reviewed; complete PDF not available in searched depots/Downloads | **Reported result:** One partial response, five stable disease, one progression. Two grade-III myelosuppression events, one treatment cessation; proteinuria in four. No rejection reported over limited follow-up.<br>**Do not infer:** No control, tiny denominator and short follow-up; does not demonstrate durable salvage, long-term graft safety, or justify enrolling G3/extrahepatic cases because PRRT is available later. |

| Study • population • source | Results • interpretation boundary |
|---|---|
| **NLRB oncology guidance** [2]<br>Official Adult MELD Exceptions for Transplant Oncology guidance, December 2025 posting checked 24 October 2026. Allocation-review guidance, not national clinical policy or proof of efficacy.<br>Primary source \| Full-text PDF<br>Locator: Official PDF NET pp.8–9; CRLM pp.9–10; introduction on purpose/status.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** No efficacy denominator. NET guidance has no explicit numerical age or hepatic-volume cutoff. CRLM wording excludes >10% diameter growth or new lesions and labels this RECIST 1.1, although conventional RECIST 1.1 progression generally uses at least 20% summed-diameter increase plus an absolute change; record both, do not silently equate them.<br>**Do not infer:** Clinical judgment, exception endorsement and graft availability are distinct. A reasonable research hypothesis or living donor does not override allocation guidance or demonstrate acceptable benefit. At exactly CEA 80 the initial/extension wording differs; the draft's strict <80 harmonization is a conservative local rule. |
| **ENETS 2026** [4]<br>ENETS 2026 position statement explicitly addressing controversies; expert, hypothesis-generating framework rather than original comparative outcome study.<br>Primary source \| Full-text PDF<br>Locator: J Neuroendocrinol 2026;38:e70 250; complete position-statement PDF burden, benefit, age, stability and extrahepatic discussion.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** No original outcome denominator. Proposed liver-burden bands around 25–70%, age <60 as relative and six-month stability are pragmatic concepts. Illustrative non-LT five-year survival <30% versus post-LT ≥70% are not a calibrated individual benefit model.<br>**Do not infer:** Do not turn an expert future-oriented proposal into validated numerical gates. Suggested controllable extrahepatic disease, including bone, conflicts with current NLRB solid-organ-metastasis exclusion. It identifies research needs rather than approving an expansion. |

## SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

Evidence supports carefully selected well-differentiated G1/G2 portal-drained GEP primaries with cleared primary/extrahepatic disease and observed control. G2 has worse prognosis than G1 in the multicentre series but still meaningful selected survival, so a universal Ki-67 <10% rule would discard potentially suitable G2 without validation. Well-differentiated G3 and poorly differentiated NEC must be separated diagnostically, yet neither has an established LT-benefit pathway. Prior primary regional nodal positivity is not equivalent to active unresected extrahepatic disease. [3] [6] [5] [14] [12] [15] [2] [4]

- Milan LT cohorts included many patients whose primary nodes had been positive, so 'ever node-positive means never LT' is too broad. However, retrospective final-pathology positive nodes do not negate the current cleared-disease requirement. Spanish OUT cases and the tiny G3 subgroup are too heterogeneous to validate expansion. NETTER-2 proves PRRT activity in selected G3, not LT efficacy after response; seven post-LT PRRT cases cannot guarantee safe salvage. ENETS controllable bone disease conflicts with NLRB's solid-organ exclusion.

**Certainty / applicability:** Low-to-moderate certainty for G1/G2 prognostic distinction and cautious standard selection; very low certainty for G3, nonportal/unknown primary, active nodes or prior solid-organ metastasis expansions. Official allocation exclusions should be reported separately from biological evidence.

## WHERE EACH PROPOSED THRESHOLD COMES FROM

- Ki-67 <20% and six months after primary/extrahepatic clearance reflect current NLRB wording; modern WHO grading and the exact-20 boundary need explicit reconciliation. Six months stability derives from selection traditions, not a randomized timing trial. Three serial scans at roughly 0/3/6 months are a proposed local demonstration schedule, not a national mandated scan count.

## SOURCE WORDING — READ IN CONTEXT

| Reference and exact excerpt | What it supports / does not establish |
|---|---|
| **NLRB oncology guidance** [2]<br>"mitotic rate less than 20 per 10 HPF and index less than 20%."<br>December 2025 oncology guidance p8, NET criterion 3<br>Source \| Full-text PDF | Retain the exact published guidance threshold and clarify pathology sampling/measurement uncertainty. |
| **Mazzaferro 2016 · Milan-NET** [3]<br>"Stable disease/response to therapies for at least 6 months prior to transplant consideration"<br>Milan-NET, Table 1, PDF p2 / journal p2893<br>Source \| Full-text PDF | A biological selection interval; the proposed three-scan workflow is not itself a validated benefit predictor. |
| **ENETS 2026** [4]<br>"solitary bone lesions amenable to local treatment"<br>2026 full-text PDF p3, section 4, proposed candidate profile<br>Source \| Full-text PDF | ENETS discusses selected controllable extrahepatic sites, including this example. Current NLRB solid-organ-metastasis exclusion is stricter. I favor no initial expansion despite this emerging clinical-consensus proposal. |

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

## OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

**Proposed stance:** Activate only well-differentiated G1/G2 disease from a known portal-drained GEP primary; retain a six-month biology test. [2] [3] [4] [6] [5] [14] [12] [15]

- Require expert pathology with Ki-67 <20% and the applicable mitotic criterion; use the highest reliable representative grade.
  - Exactly 20%, rounding uncertainty or discordance is HOLD for adjudication, not rounding down.
  - Do not impose an unsupported Ki-67 <10 rule or blanket G2 ban.
- G3, NEC, unknown/nonportal primary and active extrahepatic disease have no active cohort in this edition.

  - Previously resected nodes may qualify after complete clearance and the required disease-free interval; document stations and dates.
  - Prior lung/bone/other solid-organ metastases remain outside the current opening pathway.
- Require ≥6 months of stable/responding hepatic disease and ≥6 months without recurrence after primary/extrahepatic clearance.
  - Proposed documentation: at least three comparable assessments spanning baseline, about three months and six months.
  - A suspicious new site is HOLD; confirmed progression restarts observation.

**Why retain or change it:** Keep OPEN expert pathology review of primary and representative metastatic tissue, recording differentiation, Ki-67 hot spot, mitotic count and heterogeneity; use the highest representative confirmed grade, with rebiopsy only if clinically indicated. Advance only with known portal-drained GEP primary, well-differentiated G1/G2 and Ki-67 <20%, cleared primary and extrahepatic disease, and at least six months control/NED of resected extrahepatic sites. Ki-67 exactly 20% is HOLD for pathology/allocation clarification, not automatically renamed G3. Active G3, NEC, unknown/nonportal primary or active extrahepatic disease stays outside this pathway; prior lung/bone/solid-organ metastasis stays outside current exception guidance. Cleared regional nodes may be reconsidered after documented six months NED, not ignored. PRRT response does not waive these boundaries. Decision remains blank.

## INFORMATION REQUIRED BEFORE THE DECISION

Expert grade/Ki-67/mitoses and primary origin; SSTR staging and cross-sectional imaging; complete primary/node/EHD clearance history; three-assessment timeline.

**Clinical versus allocation:** Current NET guidance and clinical consensus differ on some boundaries; favorable guidance does not remove the need to prove B1 benefit.

**Roadmap / protocol reconciliation:** Preserve the roadmap's distinction between well-differentiated G3 NET and poorly differentiated NEC; neither gains LT eligibility from systemic/PRRT response alone.

Meeting record: NET-B2 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

## NET-B3 — SHOULD AGE OR BURDEN REMAIN LOCAL SELECTION FACTORS WITHOUT BEING CALLED NATIONAL RULES?

## QUESTION AND COMPARISON

Should age, liver tumour burden and stability be hard selection thresholds, and how should modest-burden versus >50% involvement influence the benefit decision?

## EVIDENCE BEFORE CHOOSING A RULE

| Study • population • source | Results • interpretation boundary |
|---|---|
| **Mazzaferro 2016 · Milan-NET** [3]<br>Single-centre comparative Milan-NET study, 88 eligible patients: 42 LT and 46 non-LT, 1995–2010 era, nonrandomized.<br>Primary source \| Full-text PDF<br>Locator: Am J Transplant 2016;16:2892–2902; full PDF Methods statistical time origin/left truncation, Tables 1–2 eligibility/characteristics, survival and restricted-mean analyses.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Median follow-up 122 months. Five-/ten-year OS 97.2%/88.8% LT versus 50.9%/22.4% non-LT. Time origin was primary-tumour resection; authors used left-truncated Cox models to account for delayed LT entry. Propensity-adjusted restricted mean survival gain was 6.82 months at five years and 38.43 months at ten years. These are cohort estimates, not individual calculators.<br>**Do not infer:** Major selection/age/treatment-era differences, small numbers and residual confounding remain despite statistical handling of delayed treatment. Only 4/46 non-LT patients received PRRT, so this is not a modern sequential-care comparator. Patients overlap subsequent Milan series and some multicentre LT studies; do not sum them as independent evidence. |
| **Maspero 2022 · Milan long-term** [7]<br>Retrospective Milan single-centre series, 1984–2019: 149 liver operations (53 LT, 96 LR), with 104 Milan-eligible cases analysed (48 LT, 56 LR).<br>Primary source \| Full-text PDF<br>Locator: Am J Transplant 2022 DOI10.1111/ajt.17156; complete PDF pp.2–4 methods/time origin/cohort; later survival figures, recurrence table and Discussion.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Time zero liver operation; median follow-up 158 months LT and 126 LR. Five-year OS 95.5% versus 90%; ten-year OS 93% versus 75%. Ten-year DFS 52% versus 18%; recurrence 25/48 versus 42/56 and median recurrence time 78 versus 24 months. LT recurrence was more often multisite; LR recurrence predominantly liver. Nine LT deaths included three cancer and six non-cancer deaths, including new malignancy, sepsis/lymphoproliferative and cardiovascular causes.<br>**Do not infer:** Not randomized or anatomically exchangeable; prolonged selected follow-up and differing age can favor LT. Strong overlap with 2016 Milan-NET makes it long-term/alternative-comparator information, not independent confirmation. Late recurrence and immunosuppression-related competing causes matter even with excellent survival. |

| Study • population • source | Results • interpretation boundary |
| --- | --- |
| **Eshmuminov 2023 · LT versus resection** [6]<br>International retrospective comparison from 15 centres, 1988–2021: 455 patients (225 LT, 230 liver resection); median follow-up 97 months. Propensity matching produced 96 patients per group.<br>Primary source \| Full-text PDF<br>Locator: Ann Surg 2023 DOI10.1097/SLA.0000 000 000 005 663; PDF main Results printed p.e1067/Table 4 for matched outcomes; earlier Tables 1–3 for grade and mortality; abstract on first page differs.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Use main Results p.e1067/Table 4: matched five-year OS 75% LT versus 68.3% LR, HR 0.56 (95% CI 0.35–0.90); five-year PFS 64.2% versus 14.2%, HR 0.25 (0.16–0.39), measured from operation. Ninety-day deaths 13/225 LT (5.8%) versus 3/230 LR (1.3%). LT G1 versus G2 five-year OS 92.9% versus 70%; adjusted G2 HR 2.52 (1.15–5.52). Outside-Milan matched OS HR 0.64 (0.31–1.33), not statistically conclusive.<br>**Do not infer:** Material abstract/body discrepancy: abstract reports matched five-year OS 73% versus 52.8%, whereas main Results/Table 4 report 75% versus 68.3%. The latter is used with explicit discrepancy flag; clarification would be desirable. Matching does not remove unmeasured confounding, missing pathology, era or selection bias. The nonsignificant outside-Milan estimate does not prove no benefit, and ten G3 cases do not validate G3 LT. Overlaps centre cohorts including Milan. |
| **Kjaer 2023 · eligible non-LT panNET** [8]<br>Single-centre retrospective non-LT panNET cohort, 2000–2021: 519 screened, 41 G1/G2 metastatic patients included; 16 met at least one historical LT framework (Milan 11, ENETS 8, UNOS 13, overlapping).<br>Primary source \| Full-text PDF<br>Locator: World J Surg 2023 DOI10.1007/s00 268-022-06 736-1; complete PDF pp.2–5 methods, time origin, eligibility modifications, treatments and survival tables.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Time zero was date LT criteria fulfilled (for remaining patients, initial institutional visit after primary resection with liver metastases). Mean follow-up 5.4 years. Overall 41-patient median OS 9.3 years; five-year OS 64.7% (95% CI 48.2–81.2). Milan subset five-year OS 64.9% (32.2–97.6), ENETS 85.7% (59.8–100), historical UNOS 55.4% (26.0–84.8).<br>**Do not infer:** Small retrospective selected subgroups with very wide intervals; no internal LT comparator and treatment-era limitations. It challenges the assumption that eligible stage IV NET has uniformly poor non-LT survival; it cannot prove that non-LT is superior to LT. Avoid comparing its eligibility-based survival directly with post-LT curves as if randomized. |

| Study • population • source | Results • interpretation boundary |
|---|---|
| **Boscà 2026 · Spain** [5]<br>Spanish 13-centre retrospective series of 91 NET LT recipients, 1995–2024; 41 transplanted before 2007; 71 Milan-IN and 20 Milan-OUT.<br>Primary source \| Full-text PDF<br>Locator: J Clin Med 2026;15:938; complete PDF Results, Tables 2–5, postoperative mortality/retransplantation and limitations.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** Time zero LT; median follow-up 6.5 years. Table 5: five-year OS 71.0% IN versus 58.7% OUT, ten-year 55.5% versus 32.0%, fifteen-year 52.2% versus 16.0% (OS comparison P=0.03). Five-year DFS 58.8% versus 36.3%, not statistically significant. Recurrence 52/91; ninety-day mortality 9/91 (9.9%). Nine retransplants: two primary nonfunction, six hepatic artery thrombosis, one rejection.<br>**Do not infer:** Era, missing pathology, small OUT group and composite deviations preclude validating 50–70% burden, age ≥60, G3 or extrahepatic expansion individually. Nonsignificant DFS is not equivalence. The abstract rounds OUT five-year OS differently; use Table 5 value with locator. |
| **Le Treut 2013 · ELTR** [14]<br>European Liver Transplant Registry retrospective multicentre study, 213 recipients at 35 centres, 1982–2009; 107 before 2000 and 106 thereafter. Published primary abstract reviewed.<br>Primary source \| TMC: retrieve PDF<br>Locator: Ann Surg 2013 DOI10.1097/SLA.0b013e31 828ee17c; PMID23 532 105 primary abstract; complete PDF not located.<br>Review depth: Published primary abstract reviewed; complete PDF not available in searched depots/Downloads | **Reported result:** Three-month mortality 10%. Five-year OS 52% measured from LT, versus 73% measured from liver-metastasis diagnosis; five-year DFS 30%. Post-2000 five-year OS 59%. Poor differentiation, hepatomegaly and concurrent major resection were adverse; age >45 was adverse in a later-era analysis.<br>**Do not infer:** The striking time-origin difference illustrates why survival curves cannot be compared casually. Registry era, incomplete grading and nonrandom selection limit modern applicability. Age 45 association does not establish a biological exclusion threshold. Cohort overlap with later multicentre reports is likely. |

| Study • population • source | Results • interpretation boundary |
|---|---|
| **ENETS 2026** [4]<br>ENETS 2026 position statement explicitly addressing controversies; expert, hypothesis-generating framework rather than original comparative outcome study.<br>Primary source \| Full-text PDF<br>Locator: J Neuroendocrinol 2026;38:e70 250; complete position-statement PDF burden, benefit, age, stability and extrahepatic discussion.<br>Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed | **Reported result:** No original outcome denominator. Proposed liver-burden bands around 25–70%, age <60 as relative and six-month stability are pragmatic concepts. Illustrative non-LT five-year survival <30% versus post-LT ≥70% are not a calibrated individual benefit model.<br>**Do not infer:** Do not turn an expert future-oriented proposal into validated numerical gates. Suggested controllable extrahepatic disease, including bone, conflicts with current NLRB solid-organ-metastasis exclusion. It identifies research needs rather than approving an expansion. |

## SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

The <50% burden and relative age <60 framework has the best selected-cohort lineage, not proof of a sharp treatment-benefit boundary. Registry associations with age >45 or hepatomegaly do not establish universal bans. More burden may increase need for organ replacement but also worsen prognosis; low burden may predict good post-LT survival while yielding little extra benefit. NLRB's lack of explicit age/burden limits does not validate any expansion. [3] [7] [6] [8] [5] [14] [4]

- Spanish Milan-OUT survival is lower, but OUT combines many deviations and cannot isolate >50% burden or age ≥60. Matched outside-Milan confidence intervals are wide, not proof of futility. Kjaer's non-LT survival cautions against assuming any eligible patient needs LT. ENETS 25–70% proposal is an expert benefit-oriented hypothesis, not outcome-validated proof that 70% is a biological ceiling.

**Certainty / applicability:** Low certainty for any causal age/burden threshold; moderate confidence that these variables and liver function must be measured and followed. No prospective trial validates 25%, 50%, 70% or age 60 as a universal increment-benefit boundary.

## WHERE EACH PROPOSED THRESHOLD COMES FROM

- <50% and age <60 relative originate in Milan selection; age >45 is historical prognostic association, not selection validation. ENETS 25–70% and six-month stability are pragmatic proposals. The V6 50–70% unopened stratum and >70% initial closure are local conservative design choices. Serial volumetry and trajectory matter more than a single imprecise percentage.

## SOURCE WORDING — READ IN CONTEXT

| Reference and exact excerpt | What it supports / does not establish |
|---|---|
| **Mazzaferro 2016 · Milan-NET** [3]<br>"Metastatic diffusion to <50% of the total liver volume"<br>Milan-NET, Table 1, PDF p2 / journal p2893<br>Source \| Full-text PDF | A selection criterion in the Milan framework, not a national allocation ceiling. |

| Reference and exact excerpt | What it supports / does not establish |
|---|---|
| **ENETS 2026** [4]<br>"Age <60 years may be considered a relative, not absolute, factor."<br>2026 full-text PDF p5, Table 2<br>Source \| Full-text PDF | Supports individualized benefit/fitness assessment rather than an automatic age exclusion. |

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

## OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

**Proposed stance:** Use age as a review trigger, not an absolute exclusion; retain a conservative opening burden and a separately declared expansion band. [3] [4] [7] [6] [8] [5] [14]

- Age ≥60 triggers explicit physiological-reserve, competing-mortality and benefit review; no absolute chronological-age ceiling or invented NET-specific frailty score is adopted.
- Reproducibly estimate liver involvement. <50% is the conservative Milan-derived reference; exactly 50% through 70% is a separate prospective expansion stratum, not automatically active. >70% is outside the initial local program for conservative risk control, not because published evidence proves futility.
  - The 50–70/>70 choices are local proposals, not national limits.
- Low burden, particularly <25%, defaults to non-LT care unless a documented NET-B1 benefit trigger overrides that default.
  - Do not give harmful downstaging solely to cross a percentage boundary.

**Why retain or change it:** Keep OPEN age ≥60 as an enhanced fitness/competing-risk and benefit review, not an automatic exclusion. Retain <50% estimated hepatic involvement as the initial cohort reference, with central imaging method and uncertainty documented. Proposed 50–70% cases should remain a separately designed, unopened research stratum; >70% remains outside the initial programme for conservative risk control, not because published data prove futility. Low burden <25% generally favours continued effective non-LT management unless B1 provides a compelling symptom/function/future-control rationale. Do not permit burden alone to overrule grade, extrahepatic disease or donor risk. Decision remains blank.

## INFORMATION REQUIRED BEFORE THE DECISION

Method/date of volumetric burden; functional reserve/frailty; competing illness; symptom/benefit trigger; current treatment alternatives.

**Clinical versus allocation:** NLRB guidance has no explicit age or liver-percentage ceiling; identify these as local clinical/research choices.

**Roadmap / protocol reconciliation:** Use the NET roadmap's symptom control and systemic/liver-directed options, with volumetric follow-up rather than visual percentage guesses.

Meeting record: NET-B3 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

## 3 INDIVIDUALIZED TREATMENT AND REASSESSMENT

- Review resection/ablation, somatostatin analogs, PRRT, systemic therapy and liver-directed options with NET oncology, nuclear medicine and HPB/IR. Choose according to primary site, differentiation, receptor expression, tempo, symptoms and prior exposure. Do not mandate one neoadjuvant modality to qualify for LT. [4]
- For SSTR-positive disease, PRRT may be an appropriate disease-control option. Use the current nuclear-medicine dosing and renal/marrow-protection program. There is no validated universal PRRT-to-LT washout: document marrow, renal and hepatic recovery, dosimetry and operative timing. [12]
- CAPTEM, everolimus and other systemic treatments must follow the tumor-specific oncology plan. Oncology-dose everolimus is not interchangeable with transplant immunosuppression. Record a perioperative hold and recovery plan with pharmacy; do not prescribe simultaneous unreviewed combinations.
- Proposed local assessment: cross-sectional imaging approximately every three months during treatment/observation, with symptom, nutrition, liver function and syndrome-specific biochemical review. Use RECIST 1.1 and retain the original burden/site map. Functional imaging is repeated to answer staging or biological discordance, not at an arbitrary universal frequency.

Advance to final selection only when the primary is controlled, the relevant disease-free interval is complete, there is no active extrahepatic disease, liver disease is controlled, complete non-LT clearance is not feasible, and an explicit benefit/fitness review favors LT within the intended pathway. [2] [4]

- Hold for uncertain grade, unknown primary/drainage, discordant SSTR/FDG lesions, unclear nodal clearance or insufficient observation. Progressive metastatic disease, NEC or an unapproved G3/nonportal expansion stops this proposed route. Do not translate a chemotherapy or PRRT response into automatic transplant eligibility.

## 4 ALLOCATION AND DONOR SELECTION

- The current NET route is nonstandard NLRB consideration, generally MMaT minus 3 for candidates meeting the described guidance. It is not automatic standardized priority. Submit the actual primary, histology, proliferation, resection and negative-staging record. [2] [1]
- Before recipient activation, document the intended DDLT or LDLT route, actual exception decision if applicable, expected waiting time, and a recipient-benefit assessment against contemporary non-LT care. Do not promise priority or treat DCD, an extended-criteria graft or machine perfusion as evidence that no other candidate could benefit. [1] [13]
- For LDLT, retain independent donor advocacy and donor-team assessment, freedom to withdraw, and separate consent covering uncertain recipient benefit, operative and long-term morbidity, recovery, financial burden and possible retransplantation. Do not relax donor safety criteria or make possession of a living donor a clinical eligibility requirement. [1] [13] [16]
- The living-donor literature reports low but nonzero mortality and incomplete recovery follow-up. Use current center- and graft-specific counseling; do not pool overlapping registry studies or portray donor survival as proof of complete recovery. [16]

## ALLOC-B1 — WHAT ARE WE APPROVING?

Evidence and uncertainty before the shared rule

The separation is a governance requirement, not an empirical survival cutoff. A clinically appropriate patient can lack exception eligibility, and an approved exception does not prove clinical benefit. LDLT requires an independent donor determination; a registry entry is not permission to perform experimental transplantation. [1] [2] [17] [18]

Evidence-to-threshold distinction: Six domains: local checklist architecture, not six mandatory national approvals.; 100% applicable completion: governance safeguard, not an outcome-derived percentage.; Hold pending required authorization: operational implementation of separate clinical, donor and research responsibilities.

## OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

**Proposed stance:** Require 100% completion of the applicable clinical, allocation, donor and research determinations before activation. [2] [1] [17] [18]

- Complete six separate dated determinations: clinical suitability; standardized-exception eligibility; nonstandard NLRB review status; calculated-MELD access; LDLT donor clearance; and research authorization.
  - Mark nonapplicable domains explicitly with a reason—do not manufacture six approvals for every patient.
- Any required missing, pending or adverse determination is a HOLD.
  - An exception application is not approval, a donor is not recipient benefit, and clinical approval is not study activation.
- Recheck the current policy/guidance version before each request; record actual decision and priority rather than predicted entitlement.

**Why retain or change it:** Retain 100% completion of applicable determinations with dated responsible owners and reasoned N/A entries. A required unresolved determination means HOLD, not an assumed refusal or an authorization by default.

## INFORMATION REQUIRED BEFORE THE DECISION

Signed suitability record; exact allocation route and policy version; exception outcome; donor-team status; study and institutional activation record.

**Clinical versus allocation:** Policy 9.4 permits discretionary requests where appropriate; absence of favorable diagnosis-specific guidance is not absence of a review mechanism, and a request does not promise approval.

**Roadmap / protocol reconciliation:** Use the same clinical-versus-allocation separation in every standard and high-risk pathway.

Meeting record: ALLOC-B1 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

## ALLOC-B2 — WHAT IS THE RELEVANT ORGAN OPPORTUNITY COST?

Evidence and uncertainty before the shared rule

Use pathway-level benefit from an agreed entry date, not recipient-only survival. Compare with contemporary tumor-specific care and actual local access. LDLT avoids one direct deceased-donor allocation but creates healthy-donor risk and may generate rescue graft use. Graft quality/perfusion changes do not make opportunity cost zero. [18] [1] [19] [16] [20] [21] [22]

Evidence-to-threshold distinction: Two reviewers: local independence safeguard, not validated optimum.; 6/12/18 months: sensitivity scenarios for access/dropout, not mandatory waits.; HCC ≥60% five-year ITT OS: one expert trial-design benchmark; not a cross-cancer allocation policy.; All grafts: count initial and rescue/retransplant grafts; donor morbidity remains a separate outcome.

## OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

**Proposed stance:** Require two independent recipient-benefit reviews and an explicit organ-use case; do not adopt a universal five-year survival cutoff. [21] [16] [18] [1] [19] [20] [22]

- Proposed gate: two reviewers not both responsible for recruitment must document expected incremental benefit over the actual contemporary alternative, with unresolved disagreement referred to the full committee.
- For DDLT, show local expected access/dropout and sensitivity scenarios at 6, 12 and 18 months, total graft use including retransplantation, and plausible alternative recipients.
  - These are planning scenarios, not imposed waiting periods.
- For LDLT, separately show healthy-donor risk and potential downstream deceased-donor graft use.
  - Do not call DCD, extended-criteria grafts or machine perfusion “zero opportunity cost.” No priority changes follow from an AI-selected survival threshold.

**Why retain or change it:** Retain independent benefit and organ-use reviews, but require disease-specific ITT evidence and explicit uncertainty. No universal five-year survival cutoff or numerical allocation weight is adopted.

## INFORMATION REQUIRED BEFORE THE DECISION

Same-eligibility/time-zero comparator; expected access; dropout; donor/graft type; uncertainty intervals; graft loss/retransplant estimate; written independent reviews.

**Clinical versus allocation:** Only the applicable allocation process determines priority. Favorable post-LT prognosis, evidence grade and urgency are not interchangeable measures.

**Roadmap / protocol reconciliation:** Compare against the cancer roadmap’s actual resection, ablation, HAI, radiation or systemic alternative—not historical untreated survival.

Meeting record: ALLOC-B2 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

## ALLOC-B3 — WHAT MUST DONOR CONSENT AND FOLLOW-UP INCLUDE?

Evidence and uncertainty before the shared rule

Donor mortality is low but not zero; complication, recovery and financial burdens are more frequent. Registry missingness and selected healthy donors limit reassurance. Consent must include uncertain recipient benefit,

recurrence and possible early graft loss, without pressure to proceed or relax donor criteria. [16] [23] [24] [25] [1]

Quantitative donor context: Kelly reports 8/8,619 deaths by 90 days across eras; A2ALL recorded ≥1 complication in 296/740 donors. These are different outcomes/eras, not poolable rates. The complete donor evidence and follow-up limitations are in ALLOC-B3.

Evidence-to-threshold distinction: Baseline/6 weeks/6 months/1/2 years then annually to five: proposed support schedule; current mandatory reporting remains separately checked.; One sentinel event: conservative review/pause trigger, not a statistically estimated harm boundary.; No relaxed donor criteria: protection against shifting uncertain recipient benefit onto a healthy person.

## OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

**Proposed stance:** Do not relax donor criteria; fund independent follow-up and pause after a serious sentinel donor event. [1] [16] [23] [24] [25]

- Proposed readiness gate: independent donor evaluation/advocacy, documented freedom to withdraw, recipient-benefit uncertainty consent, and funded medical, psychological, recovery and financial follow-up before investigational LDLT.
- Proposed local follow-up: baseline, six weeks, six months, one year, two years, then annually through five years.
  - Meet current OPTN reporting duties separately; this five-year support plan is not claimed to be a national reporting requirement.
- One donor death, life-threatening complication, permanent major disability or liver failure requiring transplantation triggers an immediate pause of the relevant investigational LDLT activity and independent review.
  - Other serious donor harm also warrants review; reopening needs independent authorization.

**Why retain or change it:** Retain independent advocacy, ability to withdraw, funded medical/psychological/financial follow-up and sentinel-event pause. Add explicit denominator/era caveats to consent estimates and collect local outcomes prospectively.

## INFORMATION REQUIRED BEFORE THE DECISION

Graft/resection-specific and center-specific risk; donor recovery/employment baseline; independent consent and advocate records; funded follow-up/support; sentinel-event response owner.

**Clinical versus allocation:** LDLT clearance is separate from recipient selection; the potential need for retransplantation must be included in organ-use discussions.

**Roadmap / protocol reconciliation:** The same independent donor safeguards apply whether recipient disease is standard or experimental.

Meeting record: ALLOC-B3 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

- Do not use an abstract-only mixed-malignancy LDLT survival figure to counsel a NET candidate or justify a donor. Counseling must name the relevant NET population, modern alternatives and uncertainty about incremental benefit. [3] [5] [4]

## 5 OPERATIVE EVALUATION AND SURGERY

- Plan LT only after final staging and all approvals. Inspect and sample suspicious extrahepatic or hilar nodal findings; malignant extrahepatic disease prevents proceeding under the liver-only pathway. NLRB guidance does not require exploratory laparoscopy/laparotomy solely before an exception request. [2]
- Define any further nodal sampling by actual findings and the approved surgical plan, not an automatic extended lymphadenectomy. Record explant grade, Ki-67, burden and treatment response.
- Multivisceral transplantation for mesenteric disease is outside this protocol and requires separate intestinal-transplant, evidence, donor and allocation review.

## 6 IMMUNOSUPPRESSION AND GRAFT CARE

- Follow institutional induction, corticosteroid and maintenance protocols. No NET-specific regimen has proven recurrence benefit. Consider mTOR modification only after appropriate healing/vascular assessment and pharmacy review; individualize calcineurin exposure to graft, renal, infection and oncologic risks. [13]
- Review all anticancer, hormonal and immunosuppressive drug interactions. Document graft monitoring and for-cause investigation of abnormal liver tests; do not prescribe an unvalidated routine biopsy schedule.

## 7 SURVEILLANCE

- Proposed local schedule for approval: chest/abdomen/pelvis cross-sectional imaging every 3–6 months for two years, every 6–12 months through year five, then individualized long-term assessment. NET recurrence can be late; the decision to reduce imaging should consider grade, prior tempo and treatment opportunities.
- Use syndrome-specific markers and SSTR/FDG imaging when indicated. A nonspecific biomarker elevation alone does not diagnose recurrence; interpret renal function, medication effects and clinical context.

## 8 RECURRENCE MANAGEMENT

- Confirm and restage recurrence, with tissue when needed to assess grade change. The NET/transplant MDT may consider local therapy, SSA, PRRT or systemic treatment according to extent, grade, receptor status, graft function and patient priorities. [4]
- Post-LT PRRT reports are small selected case series, not proof of a universal safe or effective salvage strategy. Review marrow, renal and graft safety explicitly. [15]

## 9 OUTCOMES AND SAFETY OVERSIGHT

- Register every referred or screened candidate in the institutionally approved system, not only recipients. Record acceptance or rejection, treatment entry, response, listing, exception decision, conversion to resection or ablation, progression, toxicity, decompensation, death, patient withdrawal and access-related loss.
- Report overall survival from the prespecified entry point and separately from listing and transplantation. Keep recurrence-free survival, recurrence incidence, graft survival and survival after recurrence distinct, with definitions, confidence intervals, numbers at risk and follow-up completeness. Do not call recipient-only survival an intention-to-treat outcome.

- Capture 90-day recipient morbidity and mortality, biopsy-defined rejection, infection, arterial and biliary complications, graft loss, retransplantation and total grafts used. Record recurrence site, salvage treatment, time off systemic therapy, functional recovery and quality of life, not survival alone.
- Before a transplanting study opens, biostatistics and the independent safety board must approve indication-specific monitoring, evaluable denominators, pause authority and reopening criteria. ALLOC-B4 now proposes explicit initial-cohort review triggers for ratification; they are not validated efficacy/futility boundaries. No universal cross-cancer five-year survival cutoff is adopted.
- Use a contemporary comparator with the same eligibility and time zero. Account for the interval needed to survive to transplantation, crossover, competing risks and overlapping study cohorts. A nonsignificant difference is not proof of equivalence.

## ALLOC-B4 — WHAT DATA ARE REQUIRED BEFORE EXPANSION?

Evidence and uncertainty before the shared rule

A small pilot can test feasibility, staging fidelity, data capture and acute safety, not prove durable survival benefit. Define time zero, all-screened enrollment, waitlist failures, post-LT survival/recurrence, competing death, all grafts, donor harms and patient-reported outcomes before opening. An independent board needs actual pause/reopening authority. [19] [26] [17] [16] [27] [28]

Zero observed events among ten fully followed independent patients still has a one-sided exact 95% binomial upper bound of 25.9% (1–0.05^(1/10)). Thus ten survivors cannot establish a <5% event risk. Ninety percent data completeness permits one missing patient; safety events and vital status require active ascertainment, not tolerance of unaccounted loss.

Evidence-to-threshold distinction: Ten per stratum and interim after five at 90 days: local feasibility cap and safety review cadence.; Twelve-month minimum: earliest expansion discussion, not proof of long-term cancer benefit.; ≥90%: proposed non-vital key-measure completeness threshold; not permission to ignore deaths/graft events or unknown status.; One early death/graft loss/re-LT: review/pause trigger, not automatic futility or withholding rescue.

## OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

**Proposed stance:** Use a small staged pilot with all-screened accounting and explicit safety pauses; no automatic expansion on favorable recipient survival. [19] [26] [17] [16] [27] [28]

- Proposed initial cap: ten transplants per separately approved biological stratum.
  - Review after the first five have 90-day follow-up; no expansion decision until all first ten have at least 12-month follow-up.
  - Reconcile vital status, graft events and known serious harms for every participant; require ≥90% completeness for other prespecified key outcomes.
- In an initial experimental cohort, one all-cause recipient death, graft loss or retransplantation within 90 days triggers a pause of further experimental enrollment/transplantation pending independent review.
  - A later possibly treatment-related death/graft loss or a donor sentinel event also triggers review/pause.
  - Rescue care is never withheld; reopening requires independent authorization.
- These are proposed conservative review triggers, not statistically validated efficacy/futility boundaries.
  - Before opening, biostatistics and the independent board must ratify or replace them with indication-specific rules, denominators, pause authority and reopening criteria.

- Capture every screened candidate, including nonrecipients, conversion to resection, dropout, deaths and access barriers; report survival from entry and from LT, recurrence, graft use, donor recovery and quality of life.
  - Ten survivors alone do not establish benefit or authorize expansion.

**Why retain or change it:** Retain the small staged pilot as an OPEN governance proposal, not an efficacy claim. Clarify that known safety/vital-status data must be reconciled for every enrolled patient; ≥90% applies only to other prespecified key measures. Indication-specific statistical design may replace the proposed numbers before opening.

## INFORMATION REQUIRED BEFORE THE DECISION

Locked eligibility strata; prospective entry/time-zero dataset; modern comparator; adverse-event definitions; follow-up-completeness report; independently signed statistical and safety plan.

**Clinical versus allocation:** Study monitoring never changes exception criteria or guarantees future access. Audit access by referral, geography, resources and donor availability without turning protected characteristics into exclusion rules.

**Roadmap / protocol reconciliation:** Retain cancer-roadmap treatment alternatives and document why non-LT care was chosen when candidates do not proceed.

Meeting record: ALLOC-B4 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

Resolve NET-B1–B3 and ALLOC-B1–B4 before changing the eligible population. Include long-term symptom control, quality of life, donor recovery and graft/retransplant use in addition to recipient survival.

Evidence review record: Five analyst comments and reconciliation | Prospective outcomes dataset

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## REVISION NOTE

Revised 2026-10-24 06:55 CDT. Evidence-before-recommendation review of all 27 B questions. Earlier protocol versions and external analyst records are preserved; this is an Analyst A synthesis, not a new external-analyst vote.

**Problem:** The earlier B proposals had an uneven evidence base and emphasized a stance before its supporting and contrary data. **Correction and reason:** Added question-level primary-study tables with selection, denominators, outcome clocks, harms, limitations, exact source excerpts and threshold provenance before each OPEN proposal. Sources/PDF links are adjacent; missing full texts remain flagged.

**Problem:** Local operational numbers could be mistaken for validated eligibility or allocation thresholds. **Correction and reason:** Separated outcome-derived estimates, guideline/policy wording and local safeguards; revised proposals when the deeper review changed their evidentiary basis. Institutional decisions remain blank.

Approval record  Clinical lead _____  Allocation review _____  Research authorization if required _____  Donor review if applicable _____  Date _____